Wellness Grocery Inventory Analysis

Week 2 Deliverable: Data Audit & Exploratory Data Analysis

Code: code/week2_data_audit.py (runs end-to-end from data/food.csv). Interactive dashboard: outputs/week2/data_audit_dashboard.html.

1. Data overview

Item	Value
Items (rows)	7,413
Columns	10: Category, Description, and 8 numeric nutrients
Units	Grams per 100 g of food (Kilocalories: kcal per 100 g)
Missing values	0
Unique categories	1,181 (588 categories contain only one item; 198 items are "No Category")
Largest categories	Beef (457), Pork (304), Ready-to-eat cereals (291), Baby food (221), Lamb (198), Soup (188)

Key takeaway: The numeric data is complete, but the **Category** label is far too fine-grained (1,181 labels) to evaluate clusters against. Before Week 5, the team should map categories into about 8–12 broad food groups, such as meat, dairy, grains, and sweets.

2. Data quality audit

Check	Rows flagged	Interpretation
Exact duplicate rows	2 (1 pair)	"BABYFOOD,MEAT,BF,STR" appears twice. Remove one copy.
Sugar > Carbohydrate	67	Physically impossible; mostly dairy items off by under 1 g (lactose measured separately). Minor measurement noise.
Fiber > Carbohydrate	2	Impossible; likely entry errors.
Kcal differs from 4P + 4C + 9F by > 100	56	Explainable: alcohol (7 kcal/g, not a column), very high-fiber cereals and cocoa, sugar-alcohol candies.
Out-of-range values	0	All values lie within 0–100 g per 100 g, and macros plus water never exceed 105 g.

All flagged rows are saved in data_quality_flags.csv and are good candidates for the Week 3 "misfit" question.

3. Distribution audit

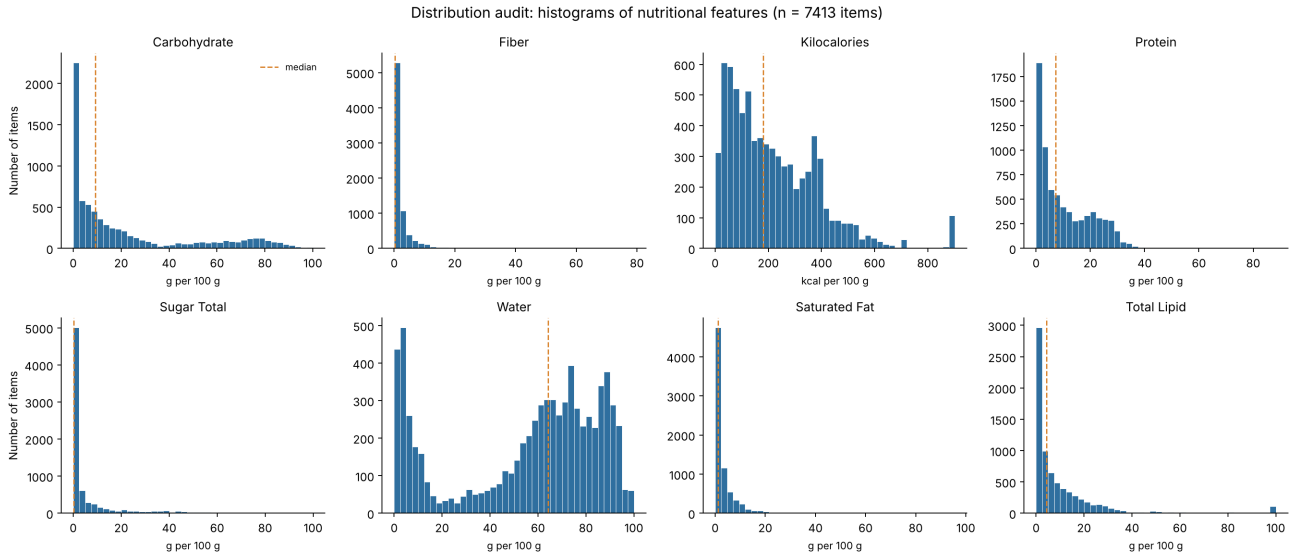


Figure 1. Histograms of the eight nutritional features (orange line = median).

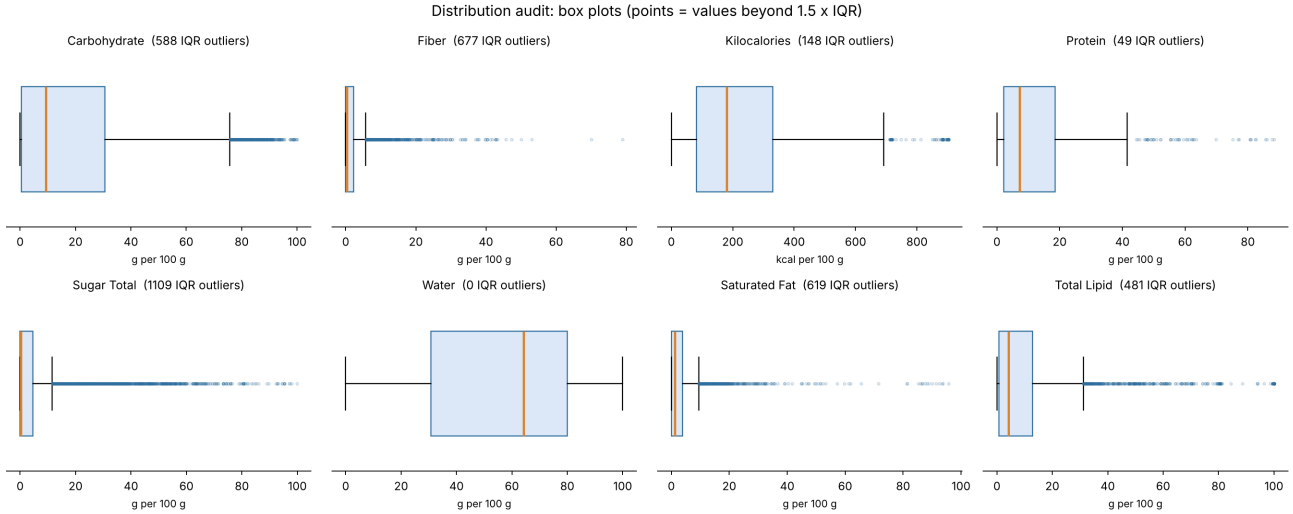


Figure 2. Box plots; points mark values beyond 1.5 × IQR.

Feature	Mean	Median	Max	Skew	% zero	IQR outliers
Carbohydrate	21.8	9.3	100.0	1.23	23%	588
Fiber	2.0	0.3	79.0	5.71	46%	677
Kilocalories	219.7	181.0	902.0	1.27	0%	148
Protein	10.8	7.3	88.3	1.38	4%	49
Sugar Total	6.4	0.2	99.9	3.05	48%	1109
Water	55.2	64.3	100.0	-0.62	1%	0
Saturated Fat	3.2	1.2	95.6	6.78	8%	619
Total Lipid	10.2	4.3	100.0	3.36	4%	481

Distribution findings

- **Most nutrients are strongly right-skewed.** Saturated fat (skew 6.8), fiber (5.7), total fat (3.4), and sugar (3.1) have most items near zero and a long tail of extreme products. Nearly half of all items contain **no fiber (46%)** or **no sugar (48%)**.
- **Water is bimodal.** There is one group of dry foods (0–15 g, e.g., cereals, snacks, oils) and one of moist foods (60–95 g, e.g., meats, produce, soups). This split is likely to become one of the main cluster boundaries.
- **Kilocalories has a spike at about 900 kcal.** These are pure oils and fats (around 100 g fat per 100 g), which also appear as the spike at 100 in Total Lipid.
- **Protein is moderately skewed** with a second hump at 20–30 g, which corresponds to meat, poultry, and fish.
- **The "outliers" are real foods, not errors.** The 1,109 sugar outliers are candies, syrups, and sweetened cereals; the fat outliers are oils, butter, and nuts. These items are exactly what a wellness audit should surface, so we keep them rather than delete them.

4. Correlation matrix: which nutrients appear together?

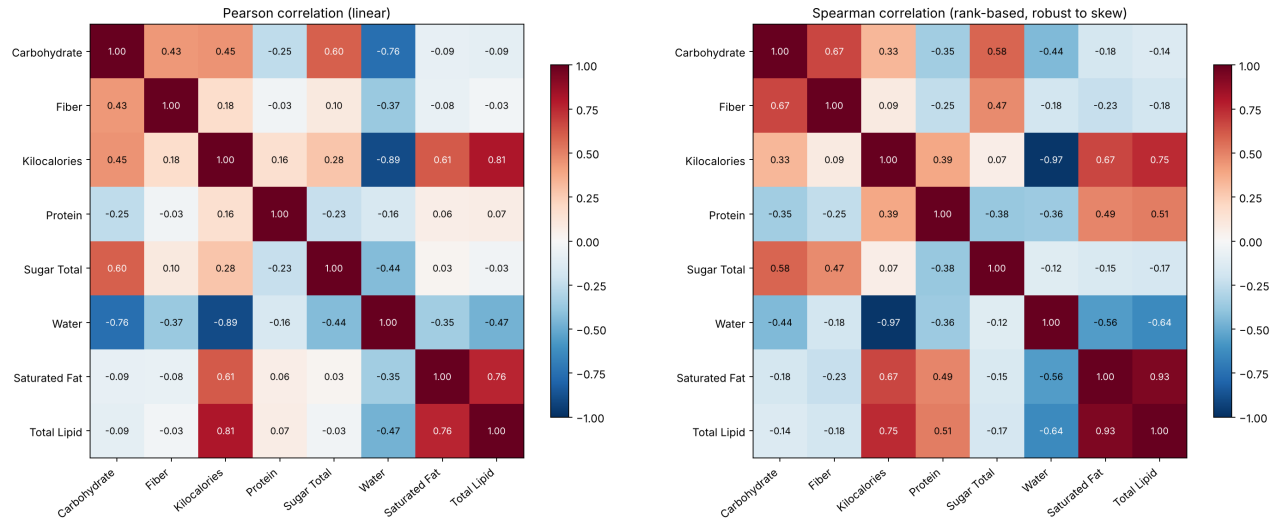


Figure 3. Pearson (left) and Spearman (right) correlations. Spearman is more reliable here because the features are skewed.

Nutrient pair	Pearson	Spearman	What it means
Water – Kilocalories	−0.89	−0.97	Water dilutes energy: moist foods are low-calorie, dry foods are calorie-dense.
Saturated fat – Total fat	0.76	0.93	Fatty foods are also high in saturated fat; the two are nearly redundant.
Total fat – Kilocalories	0.81	0.75	Fat is the main driver of calories (9 kcal/g).
Carbohydrate – Fiber	0.43	0.67	Starchy plant foods (grains, legumes) carry most of the fiber.
Carbohydrate – Sugar	0.60	0.58	Carb-heavy foods often are sugary as well.
Protein – Sugar	−0.23	−0.38	High-protein foods (meat, fish) tend to have little sugar.
Sugar – Fiber	0.10	0.47	Not "high sugar, low fiber" overall: both come from plant foods. The wellness concern is a subgroup (candies, sweet cereals) with high sugar and no fiber.

Nutrient groups

- **Energy-density group:** Kilocalories, Total fat, and Saturated fat move together, and all three move opposite to Water.
- **Carbohydrate group:** Carbohydrate, Sugar, and Fiber.
- **Protein** is fairly independent; it relates mostly to fat (Spearman 0.51), reflecting meat and dairy.

5. Implications for the modeling pipeline (Weeks 4–5)

- **Normalization is essential.** Kilocalories ranges up to 902 while fiber stays mostly below 10, so without scaling, calories would dominate k-Means distances. We recommend z-score standardization (or log(1 + x) followed by z-score for the skewed features), keeping the scaler parameters so cluster centers can be denormalized back to grams.
- **Redundancy points to PCA.** Water/Kilocalories and Saturated/Total fat are highly correlated, so PCA should compress the 8 features into a few components, likely an "energy density" axis and a "carbohydrate vs. protein" axis.
- **Category cleanup is needed** before the Week 5 purity analysis and confusion matrix (map 1,181 labels to broad groups; decide how to handle the 198 "No Category" items).
- **Data cleaning:** remove the 1 exact duplicate; keep the flagged sugar/fiber rows but note them as possible labeling errors for Week 3.

6. Team discussion: draft points for the weekly form

- What we did: loaded and audited the data (7,413 items, no missing values), built histograms, box plots, Pearson and Spearman correlation matrices, and an interactive dashboard; committed code and outputs to GitHub.
- Key findings: heavy right skew and many zeros; bimodal water; strong energy-density relationships (water vs. calories, fat vs. calories); category labels too granular.
- Decisions to make as a team: scaling method (z-score vs. min-max vs. log transform), how to group categories, whether to drop one of each highly correlated pair or rely on PCA.
- Next week: Week 3 analytical report (column meanings, 3-feature selection, outlier and misfit detection).